

Learning Guide Unit 2

Reading Assignment

As you go through the readings and watch the video, consider the following:

1. How do arrays and linked lists differ in terms of structure, memory usage, and functionality?
 2. Why might linked lists be more suitable than arrays for certain data management tasks?
 3. What challenges might arise when performing operations like insertion and deletion on linked lists, and how can these be addressed?
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How to access LIRN?

- Log into the UoPeople library and go to LIRN. Here is a link to guide you to reach LIRN - [How to Access LIRN?](#)
 - Select **Computer Science** under the **Jump to the Specific Group** section.
 - Search using the entire name of the book.
 - View the online book.
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Read

1. Cutajar, J. (2018). *Beginning Java Data Structures and Algorithms: Sharpen your problem-solving skills by learning core computer science concepts in a pain-free manner*. Packt Publishing, Limited. ([Access through LIRN](#))
 - **Read Chapter 2:** Linked Lists Structure, Linked Lists Operations (pp. 48 to pp. 54)
This chapter covers the structure and key operations of linked lists, including insertion, deletion, and traversal. It highlights their flexibility in memory usage compared to arrays and their practical advantages in software development.
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Additional Readings

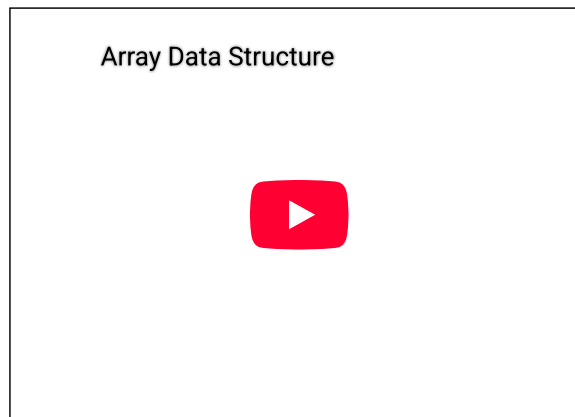
1. Morin, P. (2013). *Open Data Structures: An Introduction*. Athabasca University Press. ([Access through LIRN](#))
 - **Read Chapter 2:** Array-Based Lists (pp. 29–62)
This chapter discusses the structure and implementation of array-based lists, covering dynamic resizing, memory management, and efficiency in data storage and retrieval.
 - **Read Chapter 3:** Linked Lists (pp. 63–86)
This chapter discusses linked lists, including singly and doubly linked lists, their operations such as insertion and deletion, and their advantages over array-based lists in dynamic memory allocation.
2. Stephens, R. (2013). *Essential Algorithms: A Practical Approach to Computer Algorithms*. John Wiley & Sons, Incorporated. ([Access through LIRN](#))
 - **Chapter 3:** Linked Lists (pp. 55–82)
This chapter explores the structure and operations of linked lists, including singly and doubly linked lists. It covers insertion, deletion, and traversal techniques, highlighting their advantages over arrays in dynamic memory allocation.
 - **Chapter 4:** Arrays (pp. 83–110)
This chapter examines array structures, their fixed-size nature, and their implications for memory management. It discusses various array operations along with their efficiency and practical applications.
3. Goswami, N. (2025, January 25). *Array in data structure: Types and complexity analysis*. iQuanta. <https://www.iquanta.in/blog/a?in-data-structure-types-and-complexity-analysis/#Operations on an Array in Data Structure>

This article covers arrays in data structures, detailing their types, common operations and associated time and space complexities. It provides insights into the efficiency of arrays in computational problem-solving.

Watch

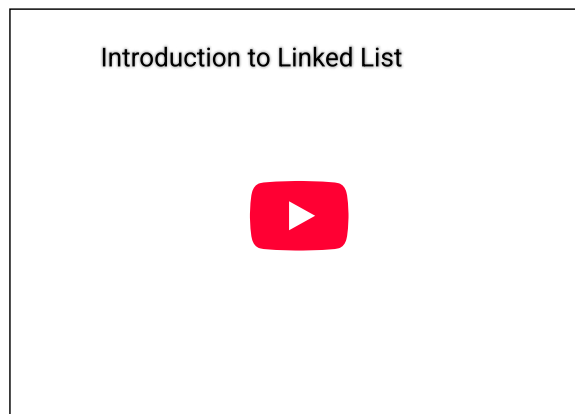
1. Telusko. (2023, August 1). *Array Data Structure* [Video]. YouTube.

This video showcases an introduction to arrays, explaining their structure, how they store data sequentially, their advantages and limitations, and common operations like insertion, deletion, and traversal.



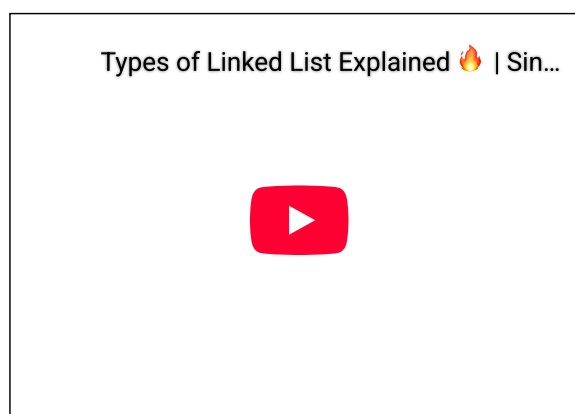
2. Neso Academy. (2020, June 22). *Introduction to Linked List* [Video]. YouTube.

This video showcases an introduction to linked lists, covering their structure, types (singly, doubly, and circular), and the key differences compared to arrays, along with practical examples of their implementation.



3. Prep Insta. (2025, March 3). *Types of Linked List Explained | Singly, Doubly & Circular Linked List | DSA for Beginners* [Video]. YouTube.

This video introduces the different types of linked lists—singly, doubly, and circular—explaining their structures, operations, and practical use cases in Data Structures and Algorithms (DSA) for beginners.



4. Bro Code. (2021, April 19). *Learn Linked Lists | LinkedList data structures and algorithms tutorial example explained* [Video]. YouTube.

This video showcases an in-depth tutorial on linked lists, explaining their structure, operations, and algorithms with practical examples to help learners understand their implementation and applications.

